# Fei Ge

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## Education

**University of California, Davis** **September 2019 – December 2024**
*Doctor of Philosophy in Physics* *Davis, California, USA*

**Xi'an Jiaotong University** **September 2014 – June 2018**
*Bachelor of Science in Physics (Honors Program)* *Xi'an, Shaanxi, China*

## Research Interests

**Cosmology and fundamental physics**: use cosmic microwave background (CMB) observations to study the evolution of Universe and the cosmological models.
**Data analysis**: develop optimal data analysis methods for low-noise CMB observations.

## Research Experience

**California Institute of Technology** **October 2025 – Present**
*Postdoctoral Scholar Research Associate* *Pasadena, CA*

**SPT-3G CMB Lensing Analysis**

- Develop the Marginal Unbiased Score Expansion method for analyzing SPT-3G CMB observations.
- Explore constraints on dark sector models from CMB power spectra inferred from SPT-3G observations.

**Stanford University** **Janurary 2025 – September 2025**
*Postdoctoral Scholar* *Stanford, CA*

**Joint Lensing Estimate of SPT-3G, ACT and *Planck***

- Develop the joint CMB lensing likelihood of joint SPT-3G, ACT and *Planck* datasets.
- Constrain cosmological parameters, including neutrino mass, using the latest lensing likelihood.

**Understand Neutrino Mass Constraints from Cosmological Data**

- Develop and modify the CAMB Boltzmann solver package to separate impacts of massive neutrinos as geometrical effects and perturbation effects.
- Use the CMB and BAO observational data to differentiate the neutrino mass constraints from the geometrical effects and perturbation effects.

**University of California, Davis** **Janurary 2020 – December 2024**
*Graduate Student Researcher* *Davis, CA*

**SPT-3G CMB Lensing Analysis**

- Built the CMB Lensing analysis pipeline using Marginal Unbiased Score Expansion method for South Pole Telescope Winter-Field CMB polarization data.
- Tested and validated the pipeline over a set of mock observations.
- Constrain and test cosmological models with the CMB bandpower likelihood.

**Atomic Dark Matter and Cosmological Tensions**

- Identified the Free-Fall, Amplitude and Thomson (FFAT) scaling transformation of cosmological perturbations that preserves the invariance of dimensionless cosmological observables, like CMB spectra.
- Applied the Atomic Dark Matter model to exploit the FFAT scaling transformation to resolve the Hubble tension problem.
- Utilized a series of scaling transformations, including the atomic dark matter (ADM) enforced one, to investigate the constraints on the amount of light relics with two of them being newly identified.
- Investigated the physical effect of changes in the ADM parameters on the cosmological observables and identified that the CMB lensing is the most impacted observable.
- Proposed the ADM model with a low dark photon temperature to be a possible model predicting low clustering strength and consistent with current dataset.

**National Astronomical Observatory of China** **July 2018 – July 2019**
*Research Assistant* *Beijing, China*

**Multi-tracer Analysis with Galaxy Surveys**

- Tested for the multi-tracer analysis with eBOSS catalog by measuring the cross- and auto-power spectrum as well as the survey window function using mock catalog.

## Fellowship and Award

Dean's Summer Graduate Fellowship *June/2023*
APS Graduate Student Travel Award *April/2022*

## Presentations

**Advancing Field-level and Simulation-based Inference for Cosmology** *June/2026*
*SPT-3G: Cosmology from CMB Lensing and Delensed EE Power Spectra Using 2019-2020 Polarization Data.*

**mm Universe 2025** *June/2025*
*SPT-3G: Cosmology from CMB Lensing and Delensed EE Power Spectra Using 2019-2020 Polarization Data.*

**15th Conference on the Intersections of Particle and Nuclear Physics** *June/2025*
*SPT-3G: Cosmology from CMB Lensing and Delensed EE Power Spectra Using 2019-2020 Polarization Data.*

**American Physical Society Global Physics Summit 2025** *March/2025*
*SPT-3G: Cosmology from CMB Lensing and Delensed EE Power Spectra with 2019 and 2020 Polarization Data*

**CMB-S4 Summer Collaboration Meeting** *July/2023*
*Atomic Dark Matter and Concordance of Cosmological Probes*

**American Physical Society April Meeting** *April/2022*
*A Symmetry of Cosmological Observables, a Mirror World Dark Sector and the Hubble Constant Tension*

## Teaching

*Teaching Assistant* at UC Davis

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|---|---|
| Introduction to Cosmology | *Spring/2023* |
| Cosmic Structure Formation (graduate level) | *Spring/2022* |
| Thermodynamics and Statistical Mechanics | *Fall/2021* |
| Introduction to Computational Physics | *Spring/2020* |
| Introduction to General Physics I | *Fall/2019* |

## Professional Activity

- Lensing Working Group Co-Coordinator of SPT-3G collaboration
- Junior member of South Pole Telescope collaboration
- LSST Dark Energy Science Collaboration member

## Technical Skills

- Python, Fortran, Julia and LaTeX
- CosmoMC, CAMB, Cobaya

## Publications

### Main Author

1. C. Trendafilova, **F. Ge**, A. R. Khalife, and C. L. Reichardt, *Cosmological Constraints on Interacting Dark Matter with SPT-3G D1*, (2026), arXiv:2610.37604

2. **F. Ge**, F. J. Qu, H. Wang, E. Schaan, W. L. K. Wu, and A. Friedland, *Disentangling geometry & growth in cosmic neutrino mass constraints: a more physical split and diagnostic*, (2026), arXiv:2610.20977

3. F. J. Qu, **F. Ge**, H. Wang, E. Schaan, W. L. K. Wu, A. Friedland, and B. D. Sherwin, *Measuring Cosmic Neutrino Masses Independently of Dark Energy*, (2026), arXiv:2607.24742

4. F. J. Qu, **F. Ge**, et al. (ACT, SPT-3G), *Unified and Consistent Structure Growth Measurements from Joint ACT, SPT, and Planck CMB Lensing*, Phys. Rev. Lett. **136**, 021001 (2026), arXiv:2504.20038

5. **F. Ge** et al. (SPT-3G), *Cosmology from CMB lensing and delensed EE power spectra using 2019–2020 SPT-3G polarization data*, Phys. Rev. D **111**, 083534 (2025), arXiv:2411.06000

6. E. Hughes, **F. Ge**, F.-Y. Cyr-Racine, L. Knox, and S. Raghunathan, *Cool dark sector, concordance, and a low $\sigma 8$*, Phys. Rev. D **109**, 103516 (2024), arXiv:2311.05678

7. **F. Ge**, F.-Y. Cyr-Racine, and L. Knox, *Scaling transformations and the origins of light relics constraints from cosmic microwave background observations*, Phys. Rev. D **107**, 023517 (2023), arXiv:2210.16335

8. F.-Y. Cyr-Racine, **F. Ge**, and L. Knox, *Symmetry of Cosmological Observables, a Mirror World Dark Sector, and the Hubble Constant*, Phys. Rev. Lett. **128**, 201301 (2022), arXiv:2107.13000

**Co-Author**

1. Y. Omori, ..., **F. Ge**, ..., et al. (SPT-3G), *SPT-3G D1: Quadratic-Estimator CMB Lensing Reconstruction and Cosmology*, (2026), arXiv:2608.31136

2. Y. Nakato, ..., **F. Ge**, ..., et al. (SPT-3G), *SPT-3G D1: Foreground-Robust Lensing Templates for Primordial Gravitational Wave Searches*, (2026), arXiv:2608.06343

3. K. Levy, ..., **F. Ge**, ..., et al. (SPT-3G), *CMB Lensing Reconstruction Using Two Years of Temperature Data from the SPT-3G Summer Survey*, (2026), arXiv:2607.05784

4. A. Ouellette, ..., **F. Ge**, ..., et al. (SPT-3G), *Cross-correlation of SPT-3G D1 CMB lensing and DES Y3 galaxy lensing*, (2026), arXiv:2606.26223

5. W. Quan, ..., **F. Ge**, ..., et al. (SPT-3G), *SPT-3G D1: Maps of the millimeter-wave sky from 2019 and 2020 observations of the SPT-3G Main field*, (2026), arXiv:2603.20163

6. S. Raghunathan, ..., **F. Ge**, ..., et al. (LSST Dark Energy Science), *Probing Physics beyond the Standard Model through Combined Analyses of Next-generation Type Ia Supernova, Cosmic Microwave Background, and Baryon Acoustic Oscillation Surveys*, Astrophys. J. **1003**, 246 (2026), arXiv:2603.09973

7. A. S. Maniyar, ..., **F. Ge**, ..., et al. (SPT-3G), *SPT-3G D1: Compton-y maps using data from the SPT-3G and Planck surveys*, Phys. Rev. D **114**, 023534 (2026), arXiv:2602.11279

8. S. Raghunathan, ..., **F. Ge**, ..., et al. (SPTpol, SPT-3G), *Measurement of the Full Shape of the Thermal Sunyaev–Zel'dovich Power Spectrum from the South Pole Telescope and Herschel–SPIRE Observations*, Astrophys. J. Lett. **1002**, L22 (2026), arXiv:2602.10107

9. A. Vitrier, ..., **F. Ge**, ..., et al. (SPT-3G), *Towards constraining cosmological parameters with SPT-3G observations of 25% of the sky*, Phys. Rev. D **114**, 043548 (2026), arXiv:2510.24669

10. A. R. Khalife, ..., **F. Ge**, ..., et al. (SPT-3G), *SPT-3G D1: Axion early dark energy with CMB experiments and DESI observations*, Phys. Rev. D **113**, 103546 (2026), arXiv:2507.23355

11. E. Camphuis, ..., **F. Ge**, ..., et al. (SPT-3G), *SPT-3G D1: CMB temperature and polarization power spectra and cosmology from 2019 and 2020 observations of the SPT-3G main field*, Phys. Rev. D **113**, 083504 (2026), arXiv:2506.20707

12. J. A. Zebrowski, ..., **F. Ge**, ..., et al. (SPT-3G), *SPT-3G D1: Constraints on inflationary gravitational waves with two years of SPT-3G data*, Phys. Rev. D **112**, 123520 (2025), arXiv:2505.02827

13. A. Coerver, ..., **F. Ge**, ..., et al. (SPT-3G), *Measurement and Modeling of Polarized Atmosphere at the South Pole with SPT-3G*, Astrophys. J. **982**, 15 (2025), arXiv:2407.20579

14. M. Doohan, ..., **F. Ge**, ..., et al., *Quantifying bias due to non-Gaussian foregrounds in an optimal reconstruction of CMB lensing and temperature power spectra*, JCAP **09**, 048 (2025), arXiv:2502.20801

15. K. Prabhu, ..., **F. Ge**, ..., et al. (SPT-3G), *Testing the ΛCDM Cosmological Model with Forthcoming Measurements of the Cosmic Microwave Background with SPT-3G*, Astrophys. J. **973**, 4 (2024), arXiv:2403.17925

16. S. Raghunathan, ..., **F. Ge**, ..., et al. (SPT-3G, SPTpol), *First Constraints on the Epoch of Reionization Using the Non-Gaussianity of the Kinematic Sunyaev-Zel'dovich Effect from the South Pole Telescope and Herschel-SPIRE Observations*, Phys. Rev. Lett. **133**, 121004 (2024), arXiv:2403.02337

17. Z. Pan, ..., **F. Ge**, ..., et al. (SPT), *A Measurement of Gravitational Lensing of the Cosmic Microwave Background Using SPT-3G 2018 Data*, Phys. Rev. D **108**, 122005 (2023), arXiv:2308.11608

18. Y. Wang, ..., **F. Ge**, ..., et al., *Extracting high-order cosmological information in galaxy surveys with power spectra*, Commun. Phys. **7**, 130 (2024), arXiv:2202.05248